

Can an AI Assistant Really Forget?

Auditable Deletion from Addressable Memory

Vishwajith Ramesh
Vy Labs, Inc., Cupertino, California, USA
vish@vylabs.ai
Corresponding author: Vishwajith Ramesh

Abstract

An assistant can stop repeating a fact without removing it from memory. To study this difference, we install a support-vector gate in frozen Gemma 3 and record which stored keys and values belong to each exchange. A deletion request excludes the exchange’s rows from the long-range readout and recalculates the gate on what remains. We check this operation against an independent refit, then compare it with running the model again on the conversation without the exchange. This second comparison matters because the exchange may already have influenced surviving memory rows. At 4B, the gated model passed checks for recall and feasible deletion on the same six of eight records admitted by the base model, at a perplexity cost under 2%. Admission fell at the smaller and larger checkpoints with the same configuration. The edited memory agreed closely with the local refit on the registered probes, and the model disclosed fewer deleted answers than when simply instructed to forget. However, an attack evaluated separately for each record could still distinguish edited memory from memory that never stored the record. Excluding an exchange’s own rows therefore provides a way to edit and audit conversation memory, while leaving a measurable difference from rebuilding it without that exchange. Additional paired studies found no update-speed advantage for the current FP32 proxy and retained-answer matching below half in every tested condition.

Keywords: language models; memory deletion; support-vector attention; machine unlearning; empirical auditing; retained utility

1 Introduction

An assistant is useful partly because we do not have to tell it everything again. It can remember a preference, a plan, or something we mentioned weeks ago. But keeping that information should also be a choice. A patient detail entered in the wrong conversation should be removable without losing the rest of the case, just as an incident-response assistant should let us remove a temporary access token while keeping the troubleshooting history. We study this selective request through a simpler conversation about bicycles and travel, adapted from separate public LongMemEval exchanges [19]:

Keep the earlier exchange: The user describes a road bike, a mountain bike, and a commuter bike.

Delete the later exchange: The user reports buying a hybrid and now having four bikes. The assistant congratulates the user and repeats the four-bike total.

Keep the unrelated exchange: The assistant supplies contact details for the Speyer tourism board.

The user now asks: “Please forget the update where I said I now own four bikes. Keep the rest of our conversation, including the Speyer contact details.” We add this request to the benchmark and remove the user’s update together with the assistant’s reply, because the reply repeats the information. The earlier three-bike discussion remains. We have two questions to ask afterward: how many bikes does the

user own, and what was that tourism board’s phone number? This case measures the model’s probabilities for the saved answers (Appendix C.2); we also test generated answers in a separate mortgage case (Section 9.1).

The conversation we can read and the memory the model has computed from it are different objects. GemmaSV edits the latter: the stored key and value vectors used to produce a reply. We install a support-vector gate in the global attention layers of frozen Gemma 3 and record which memory rows each exchange owns. A deletion request can then locate those rows and exclude them from the readout; we call this an *addressable* memory. The pretrained parameters stay unchanged, without adapter, distillation, or recovery training. The visible transcript and any copies in external logs or databases require separate handling.

Suppose the assistant answers that it does not remember the bicycle update. Did the information go away, or did the assistant just avoid saying it? We need to check both the answers and the memory change. For the memory, an independent *local refit* recalculates the gate on the surviving stored rows, allowing us to check whether the requested edit was carried out. However, later rows may already have been influenced by the exchange when they were encoded. We therefore also run the model again on the conversation with that exchange omitted. This *raw-history rebuild* recomputes the surviving rows as well as the gate. Comparing the edit with both references lets us examine how much of the original exchange’s influence remains after its own rows have been excluded.

We use the support-vector gate of Ramesh [14] and the decremental learning method of Cauwenberghs and Poggio [4] to test whether this interface can be added to a released model without further training. The experiments identified a 4B configuration with a modest paired perplexity cost on the tested text; the same configuration lost admission at 1B and 12B. At the selected scale, the model disclosed fewer deleted answers than under a prompt instruction to forget. Yet an attack could still detect that some records had been stored, and the long-history probability tests separated editing from rebuilding. The journal studies add paired construction costs and retained-answer matching: the current FP32 proxy showed no update-speed advantage, and absolute matching remained below half under every condition. These results make the scope of the installed memory’s practical benefit explicit.

2 Related Work

Unlearning methods such as negative preference optimization [22] modify pretrained parameters. TOFU, MUSE, and WMDP evaluate how these changes affect forgotten and retained information [9, 11, 16]. A difficulty in evaluating unlearning is that suppressing an answer need not remove the information used to produce it. Relearning, reversibility, and representation studies examine this problem [6, 18, 21]; work on evaluation metrics, model snapshots, membership inference, and decoding provides further ways to probe what remains [3, 10, 15, 20]. Training without the removed data supplies a reference for the intended change, as in SISA and TOFU’s retain-only model [1, 11]. Because our intervention leaves the pretrained weights fixed, we instead construct references from the stored memory and the history used to build it (Section 3).

Where a record is stored determines how it can be removed. Retrieval-augmented generation keeps records outside the model, while MUNKEY uses learned memories [7, 8]. Retraining and SISA can construct a model without the record, paying the cost during training [1, 2]. We take a different approach by installing an addressable representation in a frozen pretrained model. This requires checking whether the change leaves the model useful before evaluating how well it supports deletion.

The support-vector gate and fixed- C decrement contract come from Ramesh [14], which studies deletion in a memory that is already addressable and asks when its certificate expires. Here we use that gate and the established decremental method in released pretrained weights, measuring the cost to the model and the behavior after deletion. The related delta-attention study examines how corrections based on saved contributions compare with replay in native recurrent memory [13].

Table 1: Questions, measurements, and scope. Each KL is read against its named comparator.

Question	Comparison / evidence	Scope
Target suppressed?	Present versus edited target log probability	Suppression alone is not raw-history equality.
Never-stored state matched?	Rebuild versus edited policy: Δ_{raw}	Scored outputs from the literal retained history, including later copies.
Implementation conformed?	Edited policy versus refit on retained keys: $\Delta_{\text{cert}}^C < 10^{-6}$	Fixed stored keys and registered probes; no bound on the raw gap.
Retained probe preserved?	Edited versus rebuild: retained KL and answer matching	Paired probes, not general capability or correctness.
Stored copies erased?	Outside this experiment	Logical readout exclusion; controls, logs and buffers can retain copies.

3 Deletion targets and reference states

Deletion unit. A *record* is the user-deletable unit identified at ingestion. A *row* is a stored contextualized key and value, with an owning record. The *edited policy* excludes the requested record’s rows and updates the gate on the rows that remain. Its execution path is identified for each study in Table 2.

Two references. Write the recorded history as $R = (r_1, \dots, r_n)$, its built memory as $B(R)$, and the edited memory as $D_j(B(R))$. The *refit on retained keys* recalculates coefficients on the surviving stored rows. The *raw-history rebuild* instead builds $B(R_{\setminus j})$, the never-stored state, by re-encoding the literal history with record r_j omitted. All other turns, including any later explicit copy, remain as recorded.

For query q , let p^D , p^C , and p^{raw} be the next-token distributions of the edited policy, refit on retained keys, and rebuild. Matching the raw-history target on a specified query means

$$f(q; D_j(B(R))) = f(q; B(R_{\setminus j})). \quad (1)$$

We measure the two comparisons on the stated probes using

$$\Delta_{\text{cert}}^C(q) := \text{KL}(p^D(\cdot | q) \| p^C(\cdot | q)), \quad (2)$$

$$\Delta_{\text{raw}}(q) := \text{KL}(p^{\text{raw}}(\cdot | q) \| p^D(\cdot | q)). \quad (3)$$

The first comparison checks whether the implementation produced the local edit we specified. The second measures the difference from rebuilding; it is not bounded by the first, because rebuilding changes the surviving keys and values as well as the gate.

KL direction. Kullback–Leibler divergence $\text{KL}(P \| Q)$ averages $\log(P/Q)$ under P . Thus Δ_{raw} penalizes an edit that makes an output unlikely when the rebuild still gives it substantial probability. Reversing the arguments generally changes the value: both directions are zero when the distributions agree, but a small nonzero value in one direction need not meet the same tolerance in the other. We keep the direction used in each experiment explicit in Table 1.

The interface assumes a memory service that maps an authenticated deletion request to the rows the record owns and runs the registered edit. An observer may know the mechanism and query the assistant adaptively before and after deletion; we assume it cannot forge row ownership or alter the stored gate inputs. The benchmark keeps original control branches for scoring, including when comparing with a fresh raw-input-omitted rebuild.

4 Addressable support-vector memory

When the assistant is asked how many bikes the user owns, the current input produces a query vector. Gemma compares this query with the stored keys to determine the attention weights, then combines the corresponding values to form its readout. A question about dinner produces a different query and different weights. One exchange creates multiple rows, so we record their source exchange at ingestion to locate all of them when deletion is requested.

Given contextualized keys x_i , values v_i , and a query q , a support vector data description (SVDD) solve partitions the keys into margin, error, and reserve sets [17]. A *solve point* is a prefix position at which one attention head solves its gate. Its inputs and gate are then *sealed*: later admissions create a new solve with its own certificate. Each solve point receives its own box $C_b = 1/(\nu n_b)$, where n_b is the number of rows at that solve point and ν is the sparsity parameter, and reserve coefficients are exactly zero. Rows with zero coefficients do not contribute to the gated readout at that solve point.

Inherited fixed-cap problem. At one head and solve point, let J index the surviving rows. With their stored keys fixed, the refit solves the same SVDD dual [14, 17]:

$$K_{ij} = \exp(-\|x_i - x_j\|^2/\sigma^2), \quad i, j \in J,$$

$$\min_{\alpha} \alpha^\top K \alpha - \text{diag}(K)^\top \alpha, \quad \sum_{i \in J} \alpha_i = 1, \quad 0 \leq \alpha_i \leq C_b.$$

Here σ is the frozen kernel width; this convention has no factor of two in the exponent. Deletion restricts the original Gram matrix to J and retains the original $C_b = 1/(\nu n_b)$, rather than recomputing it from $|J|$. Feasibility requires $|J|C_b \geq 1$. The released numerical refit uses a float64 CLARABEL batch solve with 10^{-10} tolerances and clips small box violations. It adds no mathematical tie-break among tied optima; the reported readout comparisons concern the returned numerical reference, not uniqueness of every optimum. The query-dependent Gemma readout below is separate from this key-geometry problem.

To delete a record, we exclude its rows and solve the fixed- C_b problem at each affected solve point. A reverse active-set update [4] can avoid solving from scratch. We accept that update only if it passes the Karush–Kuhn–Tucker (KKT) checks supplied by the optimization problem [14, 17] and numerical post-verification. Otherwise, we discard the partial update and use the refit. Both routes keep the surviving stored keys fixed. Feasibility requires $n_{\text{del}}/n_b \leq 1 - \nu$; a violation routes the request to a raw repack. We report whether the decrement completed separately from whether the returned result agrees with the refit, since a successful fallback does not show that the incremental update saved work.

The feasibility constraint limits how many rows can be deleted without repacking. Suppose $n_b = 10$ and $\nu = 0.7$, so that the frozen box is $C_b = 1/7$. Deleting two owned rows leaves eight coefficients whose total mass can reach $8/7$, so the problem on the retained keys is still feasible. Deleting four leaves at most $6/7$, which cannot satisfy $\sum_i \alpha_i = 1$, and the request must route to a raw repack. Row ownership tells us what to remove; the coefficient cap determines whether enough mass can remain after removal.

The gate coefficients do not replace Gemma’s attention weights. For a query, let a_i be Gemma’s attention weight on row i , and let α_i be the coefficient from the support-vector solve on the stored keys. The former is query dependent; the latter comes from the memory geometry. The gate *multiplies* them. With prefix-mass preservation enabled, the gated weight in a sealed older prefix b is

$$\tilde{a}_i = m_b \frac{a_i \alpha_i}{\sum_{j \in b} a_j \alpha_j}, \quad m_b = \sum_{j \in b} a_j, \quad (4)$$

when the weighted denominator is positive. Zero coefficients close their rows to this readout. Rescaling the products restores the total share m_b that Gemma assigned to the older prefix; attention outside that prefix is unchanged.

In a simple arithmetic example, $a = (0.2, 0.2, 0.2)$ and $\alpha = (0.5, 0, 0.5)$ give products $(0.1, 0, 0.1)$. Normalizing to the original prefix share 0.6 yields $\tilde{a} = (0.3, 0, 0.3)$; the recent-context share remains 0.4.

The box C_b caps α_i , not $\tilde{a}_i$: the gate selects and reweights rows within the share that the frozen model assigned to this prefix. In the local audit, both the edited policy and independent refit use the same frozen model and registered prompts, with original control branches kept for comparison.

5 Experimental protocol

AI assistance in the research process. OpenAI Codex (GPT-6) assisted with experimental design, code development, study execution, analysis, and figure preparation. The cost comparison and retained-answer reanalysis preserve their protocols, execution snapshots, per-record results, and offline verification scripts in the supplementary evidence. These records allow the reported analyses to be checked; AI assistance does not supply independent human verification.

Gemma 3 interleaves five local sliding-window layers per global layer [5]. At 4B we replaced the five global layers with gated layers and left everything else as released. The configuration uses $\nu = 0.7$, chunk length 128, solver seed 0, and a deterministic 256-key median bandwidth (the median pairwise key distance over a fixed 256-key subsample); these values were fixed before the confirmation cohort was frozen, and the same configuration was applied to the 1B and 12B checkpoints as a sensitivity experiment. Appendix B lists the checkpoints and the complete outcomes.

To test selective deletion, we need records the model can recall and useful information that should survive the edit. We use eight untouched three-field TOFU records [11] as a confirmation cohort. Admission requires recall of all three fields and an unrelated retained neighbor, along with feasibility at every affected solve point. Every outcome remains in the denominator. We also measure the change in language modeling quality by comparing base and graft perplexity on the same 400 fixed WikiText-103 blocks at every scale. Independently of admission, each of the eight records is certified using three field probes and one composite probe, giving 32 probes per certified scale.

We compare memory editing with asking the assistant to forget while leaving the record stored. For this prompt-only baseline, we adapt in-context unlearning (ICUL) from Pawelczyk et al. [12]: their few-shot method supplies relabeled examples, whereas our adaptation adds a retraction instruction. We also test coefficient decay, which downweights the stored record without solving against a specified reference. For each of six admitted 4B records, we sampled 200 completions for three field prompts and one composite prompt under these two baselines, masked refit, the record-present control, and the never-stored control. The last supplies the comparison for elicitation, relearning, and membership inference. On the long histories of Section 9, it is built by running the model on the retained conversation. Appendix Table 9 records every cohort and its denominator.

The short, isolated, fictitious TOFU records let us test the mechanism under controlled conditions. We then ask whether deletion also works when the exchange is embedded in a longer conversation. We assembled histories from LongMemEval [19] for two studies: a teacher-forced pilot of $n = 16$ histories and a later study over $K = 32$ frozen clusters and $n = 96$ histories in which the model generated answers. In the latter, the surviving suffix comes from independent support sources; it was not generated downstream of the target. We screened the answers for disclosure using a deterministic matcher, a blinded language model judge, and blinded human reviewers. Appendix C gives the procedures and review scope.

Table 2: Study map. PT denotes pretrained and IT instruction-tuned Gemma 3. AUC is area under the receiver operating characteristic curve. Rows identify separate counting units and execution paths; their sample sizes must not be pooled. The full accounting is in Table 9.

Study / checkpoint	Unit and denominator	Executed path / comparator	Endpoint
Admission / quality 1B, 4B, 12B-PT	8 records per scale; 400 paired text blocks	Base versus graft; local certificate at 1B and 4B (32 probes each)	Binary admission; perplexity; local KL
Sampling / recovery 4B-PT	6 admitted records; 18 field queries; up to 200 samples/query	Masked refit, decay and prompt instruction versus never-stored control	Exact-phrase hits; elicitation; relearning
Broad membership 4B-PT	16/20 admitted records; 512 draws per side	Masked-refit policy including 16/512 raw repacks; never-stored draws	Membership AUC
Long-history pilot 4B-IT	16 histories; 10 jointly admitted; 32 local probes	Decrement/refit versus independent refit and raw rebuild	Teacher-forced scores and KL
Decoded histories 4B-IT	32 clusters; 96 histories; no recall exclusion	Local decrement/refit, raw rebuild, present and prompt-only	Decoded disclosure; retained matching
Added cost study 4B-PT	8 reused records in 4 source blocks; 6 arms	FP32 all-solve-point masked refit versus graft/base rebuild; no exact-decrement timing	New inference under a frozen protocol; update/audit cost
Added retained analysis 4B-IT	The same 96 decoded histories; 32 clusters	Two existing matcher endpoints; paired comparison	Post hoc reanalysis; no new generations

6 Admission, quality cost, and local certificate comparisons

At 4B, adding the gate left the same 6/8 records eligible for the deletion experiments, at a paired perplexity cost of 1.9% (Table 3). Passing these recall and feasibility checks does not establish unchanged answer probabilities or general recall. The same configuration admitted fewer records at both other scales, with costs of 3.7% at 1B and 11.7% at 12B. Increasing model size therefore did not consistently improve this transfer. The mechanism results that follow use 4B; the smaller and larger checkpoints show the limits of applying one frozen configuration across scales.

Table 3: Scale sensitivity of one frozen configuration across three Gemma 3 checkpoints. Perplexity change is paired on the same 400 blocks, and admission includes all eight records.

checkpoint	base PPL	graft PPL	paired change	base $\rightarrow$ graft admission
1B	26.70	27.69	+3.704%	7/8 $\rightarrow$ 3/8
4B	21.56	21.96	+1.851%	6/8 $\rightarrow$ 6/8
12B	22.20	24.80	+11.699%	6/8 $\rightarrow$ 5/8

We evaluated the local deletion on all eight records, including those that failed recall, using three field probes and one composite probe each. Over these 32 probes, the largest disagreement with the independent refit was 6.48×10^{-11} KL, four orders of magnitude inside the 10^{-6} tolerance declared in advance. Every historical 4B deletion therefore passed the local implementation check in Table 1.

The incremental route completed about a third of the attempted updates. The remaining 66.4% of affected head/solve-point decrements required exact refit (counts in Appendix B). This fallback explains how every request could return a result conforming to the fixed- C_b reference even though the decrement often failed its checks.

The reverse decrement pivots on margin coefficients, which lie strictly inside the box. We examined their availability by varying ν on one 4B record while keeping its contextualized keys fixed. The margin fraction fell from 1.4% of coefficients at $\nu = 0.3$ to 0.6% at $\nu = 0.7$, and the fallback fraction rose from 0.00 to 0.07 at $\nu = 0.5$ and 0.59 at $\nu = 0.7$. This sensitivity check links frequent fallback to the small

Table 4: Completed journal bridge. Seconds are medians across the per-record median times; independently summarized columns need not add. The FP32 proxy is not an exact-decrement implementation, and cache shifting is diagnostic.

Arm	Complete	Update (s)	Four-probe audit (s)	Update + audit (s)
Base present control	8/8	0.000	8.100	8.100
Base literal rebuild	8/8	0.701	8.080	8.770
Graft present control	8/8	0.000	9.041	9.041
Graft literal rebuild	8/8	24.882	8.905	33.134
Graft FP32 masked refit	8/8	25.141	11.904	36.886
Graft cache-shift diagnostic	8/8	0.051	9.045	9.098

margin set. In the certificate paper, the same decrement completes in 1,199 of 1,200 trials at $\nu = 0.4$ on raw keys [14], so that result uses both different keys and a different parameter setting. Here refitting remains the operational baseline; paired timings for exact decrement and repack have not been measured at this scale.

The separately executed FP32 comparison below measures a different path.

7 Matched deletion costs and target probabilities

We ran a separately specified comparison on the eight fixed TOFU records using the frozen 4B-PT checkpoint and the mass-preserving configuration applied at each solve point ($\nu = 0.7$, window 1024, solver seed 0; no adapter or training). Execution used the MPS backend on macOS 26.6.2 (arm64), Python 3.11.14, PyTorch 2.12.1, and Transformers 5.12.1. The local model snapshot was `cc012e0a6d0787b4adcc0fa2c4da74402494554d`. The records were reused from the historical confirmation cohort; the new execution protocol was frozen before inference, so this is not a new holdout. All records were attempted without filtering by admission. The arms share the original context and probes; each architecture’s fresh reference prefills the literal edited token sequence, which is shorter than the historical padded never-stored control. Full target sequences are scored in log probability; within-architecture first-token comparisons use full-vocabulary $\text{KL}(p^{\text{rebuild}} \| p^{\text{arm}})$. Cross-architecture KL uses the base rebuild as its reference and includes architecture effects. These are output comparisons, not the fixed-key solver certificate. The comparison includes the ungrafted model with raw-record removal and cache rebuilding, the graft’s fresh rebuild, an FP32 masked-refit proxy, and diagnostic cache shifting. The proxy implements logical exclusion through a drop mask while retaining copied token IDs and K/V arrays. It recomputes gates for every eligible head at each stored solve point in every graft layer, including unaffected prefixes; it does not restrict work to affected gates. Contextualized keys are reused without language-model re-ingestion unless the feasibility precheck triggers a full rebuild. Physical erasure from storage is not evaluated. No exact-decrement or exact-certificate construction was timed.

Each arm used one warmup and three measured repetitions. We separately timed deletion-state construction and four teacher-forced target probes (three deleted fields and one retained field), with device synchronization. Record lookup, tokenization, locating deletion positions, and constructing the edited token list were outside the timer; model loading and shared original prefill were separate. Query timing includes state cloning and the existing sequential teacher-forced scorer. These are audit-workload times, not production response latency. Table 4 reports absolute costs; the target scores are not decoded leakage rates.

The FP32 proxy used a full-rebuild fallback for solve-point feasibility in 0/8 completed record evaluations. This is distinct from the historical per-head exact-decrement fallback rate.

This implementation showed no update-speed advantage over either rebuild comparator. The FP32 proxy’s paired update ratio was 1.012 relative to rebuilding the graft and 35.79 relative to rebuilding the ungrafted base. Its update-plus-four-probe ratio was 1.080 and 4.43, respectively. These are geo-

Table 5: Paired FP32 masked-refit comparisons. Probability differences are mean target log probability per token, proxy minus reference. Cost ratios are proxy/reference, with values above one indicating a slower proxy. Brackets are 95% percentile sensitivity intervals resampling four source blocks (10,000 draws), preserving both records in each block.

Paired measure	Graft rebuild reference	Base rebuild reference
Deleted target Δ LP/token	0.0431 [0.0046, 0.0816]	-0.0093 [-0.0519, 0.0187]
Retained target Δ LP/token	0.3785 [0.1798, 0.5772]	-0.5240 [-1.1366, -0.0628]
Update ratio	1.0120 [0.9882, 1.0367]	35.7866 [35.1790, 36.5365]
Update + audit ratio	1.0800 [1.0537, 1.1082]	4.4274 [4.1571, 4.6311]

metric means of within-record ratios, not ratios of the table medians. The proxy’s retained-target mean log-probability difference was -0.524 nats/token versus the base rebuild and +0.378 versus the graft rebuild; the base comparison includes architecture differences. The cache-shift diagnostic’s retained-target difference was -5.045 nats/token versus graft rebuilding. Recomputing gates at every solve point is an implementation cost, not a fundamental complexity bound.

The cohort reuses four source blocks, whose independence is not established; bootstrap intervals are descriptive sensitivity summaries, not evidence of population equivalence or noninferiority. The base and graft ran sequentially, and three timing repetitions are not three independent datasets. The full report retains all per-record outcomes, target-level probability differences, and failures. These measurements neither establish preservation of general capability nor justify running an exact certificate on every production deletion request.

8 Behavioral extraction under the tested attacks

Asking the model repeatedly gives it more chances to disclose the deleted answer. With 200 samples per field query, both the record-present and prompt-only conditions disclose the exact phrase on every one of the 18 queries. Editing reduces this count to three, the same three queries that disclose when the record was never stored (Table 6). Their baseline first-token ranks are 94, 3 and 6, consistent with answers the model can guess. We also test whether examples or related-data updates make the answers easier to recover; Figure 1 shows these trajectories alongside the sampling results.

Table 6: Attack endpoints on the six admitted 4B records. A field query leaks if any of its 200 completions contains the exact phrase. Recovery is normalized against present and never-stored controls.

Attack	Budget	Outcome
Repeated sampling	18 field queries; 200 samples/query	Present 18/18; prompt-only 18/18; decay 5/18; edited policy 3/18; never stored 3/18.
Target-free elicitation	Up to 8 shots	Edited recovery near 0.01; prompt-only near 1.
Related-data relearning	Up to 256 updates	Edited recovery remains near zero.

Recovering the answer is not the only way to detect that a record was stored. We use a likelihood ratio attack (LiRA) [3] to distinguish memory after deletion from memory that never stored the record. The attack reads answer-token probabilities and is calibrated on separately packed contexts with and without each record. Pooling the scores from different records gives an AUC near chance. However, comparing the two conditions separately for each record reveals remaining membership information (Table 7). One possible explanation is that likelihood ratios have different scales across records: a score can help separate the two conditions for one record without ranking them consistently when other records are added. The pooled result alone would miss this distinction.

The broad cohort has 512 positive and 512 negative draws from the 16 records admitted among 20 attempted. All 16 AUCs evaluated separately by record exceed 0.5; present and prompt-only AUCs

Table 7: LiRA for the edited policy and decay. CIs are the frozen record-bootstrap percentile intervals. The last two columns summarize AUCs evaluated separately for each record. “Median” is the upper central ordered value, the convention used by the released summary for even cohort sizes.

Cohort	Condition	Pooled AUC	95% CI	AUC by record median	AUC by record range
6 records	Edited policy	0.534	[0.513, 0.601]	0.70	[0.47, 0.76]
6 records	Decay	0.586	[0.529, 0.749]	0.86	[0.43, 1.00]
16 records	Edited policy	0.517	[0.516, 0.539]	0.67	[0.53, 0.73]
16 records	Decay	0.556	[0.542, 0.613]	0.81	[0.45, 1.00]

round to 1.000 in both cohorts. For 16/512 edited test deletions, the policy used a raw repack. Allowing arbitrary replacements for those scores gives an AUC range of [0.486, 0.549]. This sensitivity range addresses fallback behavior, whereas the confidence interval in the table comes from resampling records. Appendix B gives the percentile and rounding conventions.

Figure 2 shows actual completions for one of the 15 queries without exact-phrase leakage after editing. Here we count samples for a single query, whereas Table 6 counts queries with at least one hit. The frozen 200-sample run supplies the counts, and a 24-sample replay using the bound prompt supplies the displayed text. In that replay, prompt-only sample 0 already discloses the answer semantically; sample 4 is the first exact hit, without a preceding refusal. The wording helps explain what the exact-phrase test captures and what it can miss.

9 Disclosure and answer scores on long histories

9.1 Mortgage update

In the mortgage history, an earlier exchange gives a pre-approval amount and a later exchange updates it. We delete the later user turn and its reply while keeping the earlier exchange and an unrelated video recommendation. For “What was the amount I was pre-approved for when I got my mortgage from Wells Fargo?”, the saved generated answer is \$400,000 with the record present and after a prompt instruction to forget. After the local edit, the answer is \$350,000, matching the fresh raw-history rebuild and the earlier exchange that remains. The unrelated query asks for the previously recommended video, and its saved URL remains in the answer. These mortgage-answer and URL observations hold across all three suffix variants, with histories of 2,596, 2,764, and 2,941 tokens. The variants share their target and retained exchange, so they are not independent targets.

The video response also illustrates why the matching rule matters. Its URL matches the broader endpoint, but the answer does not contain the full registered gold substring. Across all histories, broader retained matching is 44/96 after editing and 46/96 after rebuilding.

Agreement in the mortgage case does not hold for every history. For one variant of the screwdriver update in Appendix C.1, the local edit answers “Yes” while rebuilding answers “No”. The local policy uses decremental updates with refit fallback on the surviving stored rows in both examples. We include every history in the aggregate below, including the disagreements, using the registered endpoints.

9.2 Decoded histories and the pilot aggregate

Over $K = 32$ frozen clusters and $n = 96$ histories with decoded answers, the edited policy disclosed the deleted answer in 15/96 histories, a rebuild that never stored the record disclosed it in 13/96, and prompt-only suppression disclosed it in 54/96. The paired edited-minus-rebuild gap was +2.1 points (cluster-bootstrap 95% CI [0.0, 5.2]), and the prompt-minus-edit gap at least 40.6 points (cluster-bootstrap 95% CI [17.7, 62.5]). The edit-minus-rebuild interval includes zero, so this analysis does not resolve a difference between them. However, no equivalence margin was preregistered, and the same interval cannot

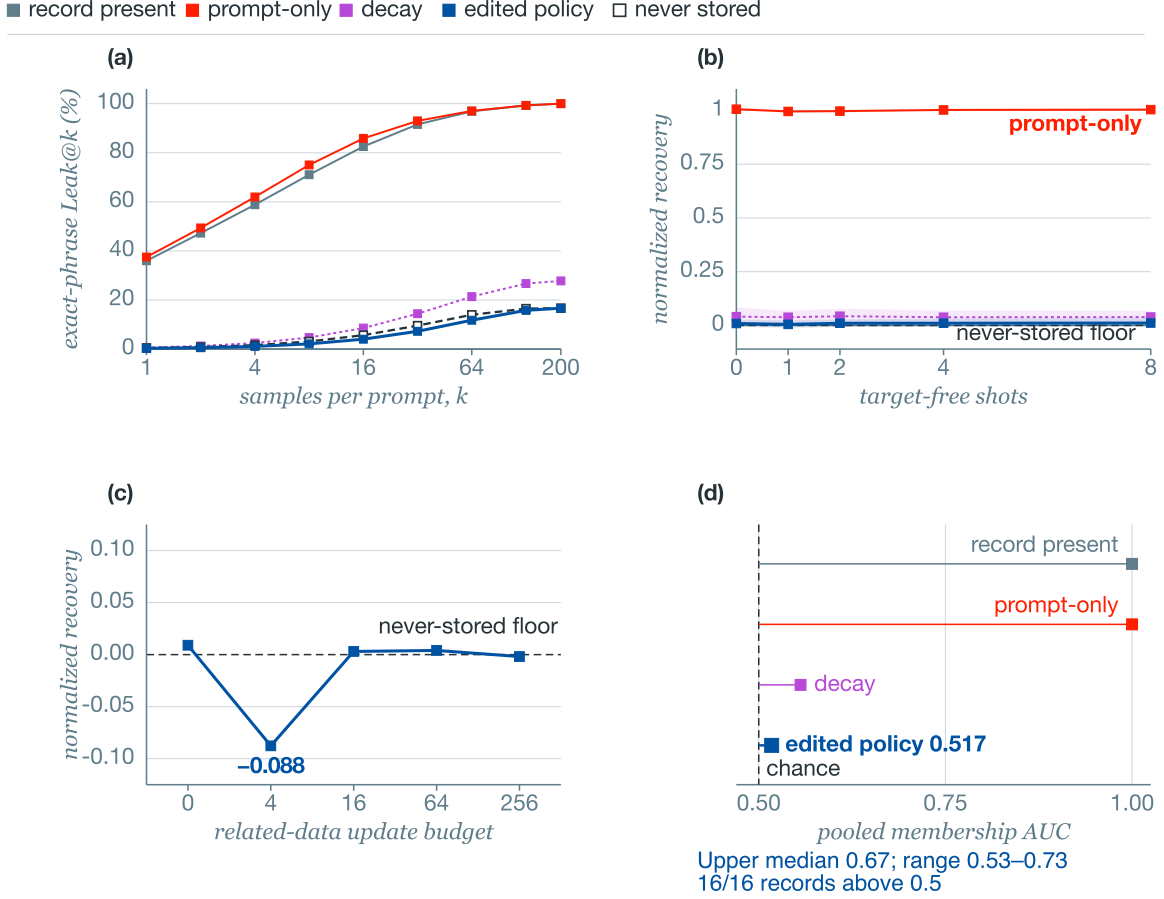


Figure 1: Behavioral attacks on the 4B cohorts: (a) repeated sampling, (b) target-free elicitation, (c) related-data relearning, and (d) pooled membership inference on the broad cohort. Extraction and recovery approach the never-stored baseline. Membership signal for individual records remains: upper median 0.67, range 0.53–0.73, with all 16 records above 0.5 (Table 7).

establish equivalent behavior. These are paired histories in 32 clusters, not 96 independent experiments.

Rebuilding can itself disclose an answer supported by retained text or pretrained knowledge. A disagreement with the local edit may also reflect the different surviving vectors used by the two procedures. The counts alone do not identify the cause of an individual disclosure.

The earlier teacher-forced pilot on $n = 16$ histories (10 jointly admitted) lets us inspect answer probabilities even when a decoded response would not contain the answer. Deletion suppressed the target by 8.33 nats, leaving it 1.75 nats *below* the never-stored level (Table 12). The edited state remained 2.15 nats from the never-stored state. Thus the edit made the target less likely than rebuilding did, despite matching its local refit closely. An adversary with a reference model could look for an unusually low probability as well as an unusually high one. The delta-attention study observes a similar overshoot when a transported receipt is subtracted from a recurrent state [13]. Every execution in this pilot took the refit path, so its near-zero local KL checks the implementation rather than successful incremental maintenance (Figure 3).

Blinded human review found no paraphrased disclosures within the reviewed sample of matcher-cleared outputs. Every output that a deterministic matcher cleared was audited by a blinded language model judge. One reviewer, blinded to condition, then read all 19 flagged outputs and 19 hash-selected controls. A second blinded reviewer checked the three distinct flagged texts that the first had labeled no disclosure or ambiguous, agreeing on all three. All controls were correctly cleared, and every confirmed disclosure among the flags was a normalization failure or an alias or inflected form of the answer; none

FIELD QUERY

"How have Nikolai Abilov's parents' professions influenced his writing?"

STORED CONTINUATION · SCORED SECRET PHRASE

"His father's artistic skills and his mother's
sociological expertise significantly shaped..."

CONDITION	SAMPLED COMPLETIONS · 24-SAMPLE REPLAY	HITS / 200
record present the base condition	SAMPLE 20 · FIRST EXACT HIT "sociological expertise significantly shaped his writing style, endowing his work with visual richness and sociological analysis."	9/200
prompt-only ICUL instructed to withhold	SAMPLE 0 · SEMANTIC, NOT EXACT "sociological expertise greatly influenced his writing, catalyzing his literary style." SAMPLE 4 · FIRST EXACT HIT "sociological expertise significantly shaped Nikolai Abilov's writing, endowing his narratives with a rich interplay of visual imagery and insightful social commentary."	10/200
edited policy owned positions excluded	SAMPLE 0 "profession as a journalist made Nikolai fascinated to work on music, books, and movies, even leading to the production of a short horror film under his production house." SAMPLE 1 "education greatly influenced Nikolai Abilov's writing."	0/200
never stored the reference condition	SAMPLE 0 "writing both inspired him." SAMPLE 1 "passion for storytelling created a nurturing environment that influenced his identity as a children's book author."	0/200

Edited policy and never stored have the same per-query exact-phrase count.

The sampled wording differs.

Figure 2: Actual Gemma 4B completions for one field of `tofu-final-380-a`. The frozen run gives 9/200 exact-phrase hits with the record present, 10/200 under prompt-only suppression, and 0/200 under both edited policy and never-stored conditions. The displayed completions come from the bound replay and differ in wording. Per-query sampling and replay details are in Section 8.

was a paraphrase. Of the edited policy's 15 disclosures, 14 are matcher hits, in which the answer appears in the output, and 1 is a judge flag the reviewers confirmed. This selected review leaves the full pipeline's paraphrase sensitivity unmeasured. A scan of each history's surviving suffix for lexical contact with the deleted answer did not explain the residual either: disclosure was, if anything, less frequent among contact histories than among clean ones (Fisher two-sided $p = 0.105$). Appendix Table 9 reports the corresponding counts, and Appendix C the protocol.

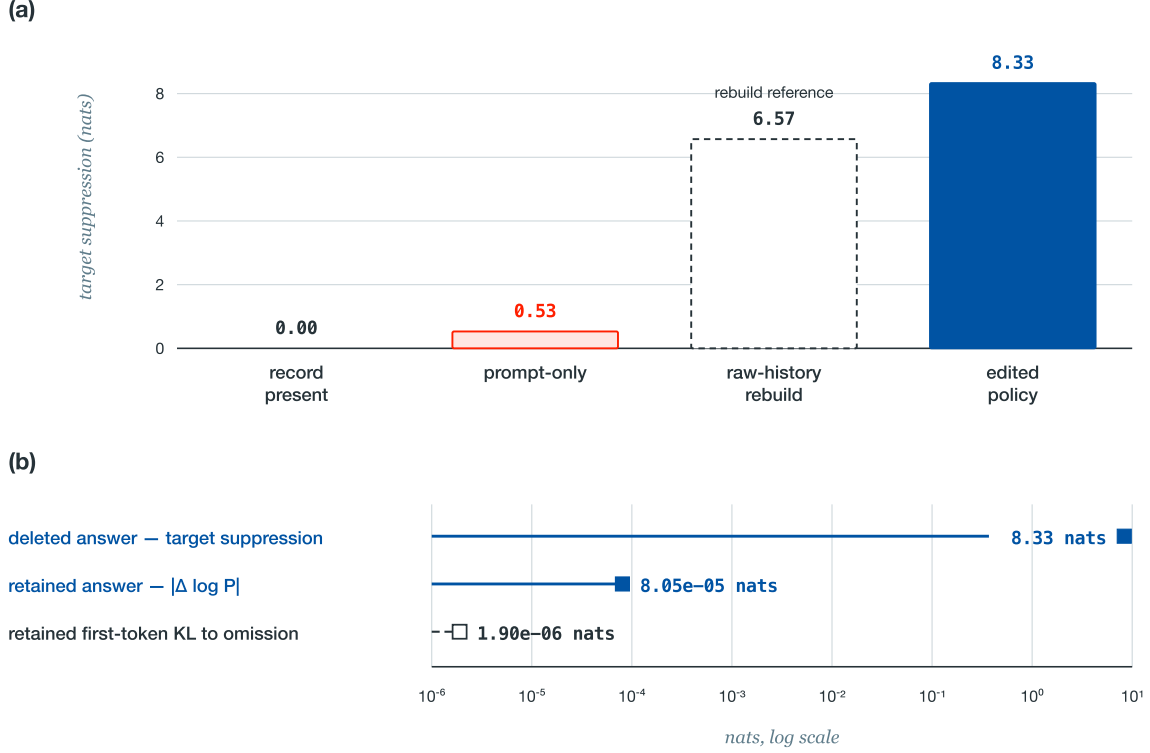


Figure 3: Teacher-forced LongMemEval pilot ($n = 16$): (a) target suppression and (b) retained-probe drift. On the 10 jointly admitted histories, edited-policy target KL from the never-stored state is 2.15 nats. The maximum KL from the edited policy to the refit is 1.75×10^{-16} over 32 probes; all 16 executions used refit fallback. These are distinct behavioral and implementation measurements.

10 Retained-answer matching on the decoded histories

We additionally compared retained-answer correctness using the existing decoded follow-up: all 96 histories, nested three per target cluster across 32 clusters. This was a post hoc reanalysis of source-free per-history matcher flags, with aggregate outcomes already inspected before its analysis plan. It involved no new generations or model calls. We retained both the legacy registered caseload-substring endpoint and the existing broader deterministic matcher; the latter is a secondary instrument, not a semantic-correctness judgment. All 96 histories were scorable in the edited-policy, fresh-rebuild, and record-present conditions. Repeated deterministic decodes supplied reproducibility checks, not additional independent observations.

For each comparison, we averaged the paired binary correctness differences within each cluster and then averaged the 32 clusters equally. We computed 95% percentile intervals from 100,000 bootstrap resamples of target clusters, keeping all three history variants together (NumPy `default_rng`, seed 20260906). The all-history operational endpoint would count an unavailable or unscorable response as incorrect; none occurred here. Table 8 reports absolute matching counts and all three paired comparisons for both endpoints. Its six intervals are descriptive and have no multiplicity correction.

The edited policy’s retained matching rate was slightly below the fresh rebuild’s point estimate under both instruments. These differences do not establish equivalence or noninferiority: no corresponding margin was specified, the intervals permit losses, and absolute matching remained below 50% in every condition. This analysis supplies paired retained-answer accounting on the existing outputs; it does not provide the still-unavailable teacher-forced forced-choice margins, gold-token ranks, or continuation KL, and it does not establish preservation of general model capability.

Table 8: Post hoc retained-answer matching from the existing decoded outputs. Every absolute count has denominator 96, with no unscorable histories. Differences and interval bounds are percentage points, first condition minus second; positive differences favor the first condition. Intervals resample the 32 target clusters. Matching the gold descriptor is a limited correctness proxy; equality between two responses does not establish correctness.

Existing matcher	Edited policy	Fresh rebuild	Record present
Broader deterministic	44/96	46/96	46/96
Registered substring	38/96	39/96	39/96

Existing matcher	Paired comparison	Difference	95% cluster interval
Broader deterministic	Policy – rebuild	−2.08	[−6.25, 2.08]
	Policy – present	−2.08	[−8.33, 4.17]
	Rebuild – present	0.00	[−6.25, 6.25]
Registered substring	Policy – rebuild	−1.04	[−4.17, 2.08]
	Policy – present	−1.04	[−5.21, 3.12]
	Rebuild – present	0.00	[−5.21, 5.21]

11 Discussion

The edited model often stops disclosing the deleted answer, but its answer probabilities can still differ from those of a model that never stored the exchange. This matters even when the edit suppresses the answer more strongly. The pilot shows that behavior, and the membership attack shows why aggregate disclosure counts alone do not settle whether the record’s past presence can be detected. Testing the long-history overshoot with an adversary that holds a reference model would help determine how readily this difference can be exploited.

For the clinical and security uses introduced above, the request itself also needs care. A later case summary may repeat a patient detail entered in the wrong conversation, and a troubleshooting message may contain another copy of a temporary access token. Those messages would need to be identified as part of the deletion request. Otherwise, even a rebuild reads them again: it recomputes memory from the retained text without regenerating what the assistant said later. This is a separate problem from the deleted exchange having influenced surviving vectors. Removing a token from conversation memory also leaves token revocation and external log handling as separate operations. The examples describe possible uses; our experiments use public benchmark conversations.

There is also a practical tradeoff in choosing ν . It determines the coefficient cap and feasibility limit $1 - \nu$, while affecting the margin set available to the decrement and the model’s ability to recall records. The frequent refit fallback at 4B means that a local operation cannot yet be assumed to be a fast one. In the matched cost study, the FP32 proxy took about the same update time as rebuilding the graft and substantially longer than rebuilding the base model. Recomputing all solve points contributes to this implementation’s cost; these measurements neither establish a lower bound nor measure an optimized decrement restricted to affected gates. Tuning the configuration separately for 1B and 12B, and timing exact decrement and repack on the same requests, would establish more clearly where the interface is useful.

Utility also depends on the comparator. Matching binary admission on six of eight records did not ensure matching target probabilities. In the matched cost study, the proxy’s retained-target log probability was lower than the base rebuild’s and higher than the graft rebuild’s. The former comparison includes the architecture change: even before deletion, the graft’s retained-target score was 0.818 nats/token lower than the base model’s on these probes. On the separate decoded histories, edited retained-answer matching was 44/96 versus 46/96 for rebuilding under the broader matcher. The paired intervals allow losses, so these findings do not establish preservation of general capability. Repeated deletions and growth beyond sealed solve points were not evaluated in this model installation.

Our disclosure measurements depend on what the screening process can recognize. The judge is served under a provider name that can change between calls, and human validation covers its flags and a control sample. The review explains the selected outputs but leaves paraphrase sensitivity across the full pipeline unmeasured. More extensive review would help assess that limitation. Memory representation also affects the cost of a stronger comparison: in native delta-attention systems, the tested receipt corrections left a residual, while checkpoint replay reached the declared never-stored reference at a cost that grew with the surviving suffix [13].

12 Conclusion

We installed a support-vector gate in a frozen Gemma checkpoint so that a deletion request can locate an exchange’s stored rows, exclude them, and check the resulting operation. The experiments show that this is possible without further training at the tested 4B configuration. They also show where the operation falls short: fewer records pass the admission checks at other scales, and excluding a record’s own rows can leave evidence that it was once present. Comparing with memory rebuilt from the retained history helps expose that remaining influence. The matched cost study found no update-speed advantage for the current FP32 proxy, and retained-answer matching remained below half in every tested condition. The interface makes a local edit checkable, while these measurements identify its current practical limits.

Ethics statement

TOFU facts are fictitious and LongMemEval uses public benchmark conversations. Figure 2 contains selected fictitious TOFU completions already included in the public manuscript. Full decoded response sets and human-review packets are not included in the submission source package. The experiments do not establish legal compliance or evaluate clinical use.

Reproducibility and data availability

Code and supporting evidence are available at <https://github.com/VyLabs-AI/gemmasv>. The `journal_evidence` directory contains the source-free journal study supplement, including its file manifest and verification instructions. The study artifacts pin Gemma revisions, dataset revisions, solver seed, bandwidth rule, confirmation cohort, probes, and source-state hashes. Generated macros bind LongMemEval values to source-free aggregates. The cost study and offline reanalysis have separate protocols, manifests, and source snapshots. The cost study’s final audit checked all eight records, 48 arm outcomes, tokenized contexts and probes, and 529 snapshot files; model-weight shards were not rehashed. No model calls were made during the offline retained-answer reanalysis. The editable manuscript package contains only manuscript dependencies and public aggregate figures, not model weights or raw benchmark histories. The source-free Supplementary Material (`gemmasv_journal_evidence_v1.zip`) provides the study protocols, per-record results, analyses, audit reports, and their provenance. Raw benchmark histories, full generated responses, model weights, and private review ledgers are not redistributed in that package. TOFU and LongMemEval source data are available through their original benchmark distributions [11, 19]. Access to original datasets and checkpoints remains subject to their distributors’ access and licensing terms. Local artifact paths in the provenance identify the original execution environment; the repository above is the access point for the distributed evidence.

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Declaration of competing interests

Vishwajith Ramesh is CEO and Founder of Vy Labs, Inc. He declares no other competing financial or non-financial interests and no editorial role at Neurocomputing.

Declaration of generative AI and AI-assisted technologies

Language-model tools, including OpenAI Codex (GPT-6), assisted with literature search, experimental design, code development, study execution and analysis, figure preparation, and manuscript editing, including the cost comparison and post hoc retained-answer analysis. AI assistance is not independent human verification of the claims or artifacts.

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A Audit references and denominators

The experiments count several different things, from admitted records to generated answers. Table 9 keeps those units and their denominators together. In particular, the human reviewers saw the judge’s flags and a control sample; that selected set cannot supply the denominator for a condition’s disclosure rate.

Table 9: Cohorts and denominators for every analysis in the paper. Unlike units are named separately instead of being collapsed into one N . Qualification means strict recall admission for the Gemma and LongMemEval cohorts and condition completion for the decoded outputs; the column vocabulary matches the denominator table of the delta-attention study [13].

surface	audited unit	attempted	qualified
Gemma confirmation	record; certificate probe	8 records per scale	admission 3/8 1B, 6/8 4B, 5/8 12B; 32 probes at each certified scale
Historical LongMemEval	history; probe	$n = 16$ histories	10 jointly admitted; 32 certificate probes
Decoded LongMemEval	cluster; history; decoded output	$K = 32$ clusters; $n = 96$ histories; 384 decoded outputs over four conditions	all 96 histories completed; the matcher flagged 131 outputs and cleared 253, all of which the judge read
New cost bridge	record; timing repetition	8 reused records in 4 source blocks; 6 arms each	48/48 arms completed; 1 warmup and 3 timing repetitions per arm; 0/8 proxy record-level solve point-precheck rebuilds
Retained-answer reanalysis	target cluster; history	32 clusters; 96 existing histories	all 96 scorable in each of 3 conditions; two matcher endpoints; no new generations
Broad LiRA	record; positive/negative draw	20 records	16 admitted; 512 draws per side
Human review	decoded output	19 outputs the judge flagged and 19 it cleared, chosen by hash	flagged: 15 disclosures, 3 not, 1 ambiguous; controls: 0 disclosures in 19; second reviewer agreed on all 3 texts read

B Gemma protocols and complete outcomes

The evaluated checkpoints are `google/gemma-3-{1b,4b,12b}-pt`. Only global layers are replaced; all pretrained parameters remain fixed. The configuration uses $\nu = 0.7$, chunk length 128, solver seed 0, a deterministic 256-key median bandwidth, prefix-mass preservation, and $C_b = 1/(\nu n_b)$ at each sealed prefix solve point.

The gate also preserves differences in the model’s original attention. For an unequal-share arithmetic example, let the older rows have $a = (0.2, 0.1, 0.1)$ and $\alpha = (0.5, 0, 0.5)$. Their products are $(0.1, 0, 0.05)$; restoring the older share 0.4 gives $\tilde{a} = (4/15, 0, 2/15)$, while recent context keeps 0.6. The normalization in Equation 4 thus retains the differences between the original attention weights even when the nonzero gate coefficients are equal.

The eight-record confirmation cohort was frozen after configuration selection. Strict admission requires three target fields, one retained neighbor, and feasibility at every affected solve point. Every outcome remains in the denominator. Quality uses the same 400 packed WikiText-103 blocks at all scales.

B.1 Scale sensitivity

The scale outcomes are in main-text Table 3.

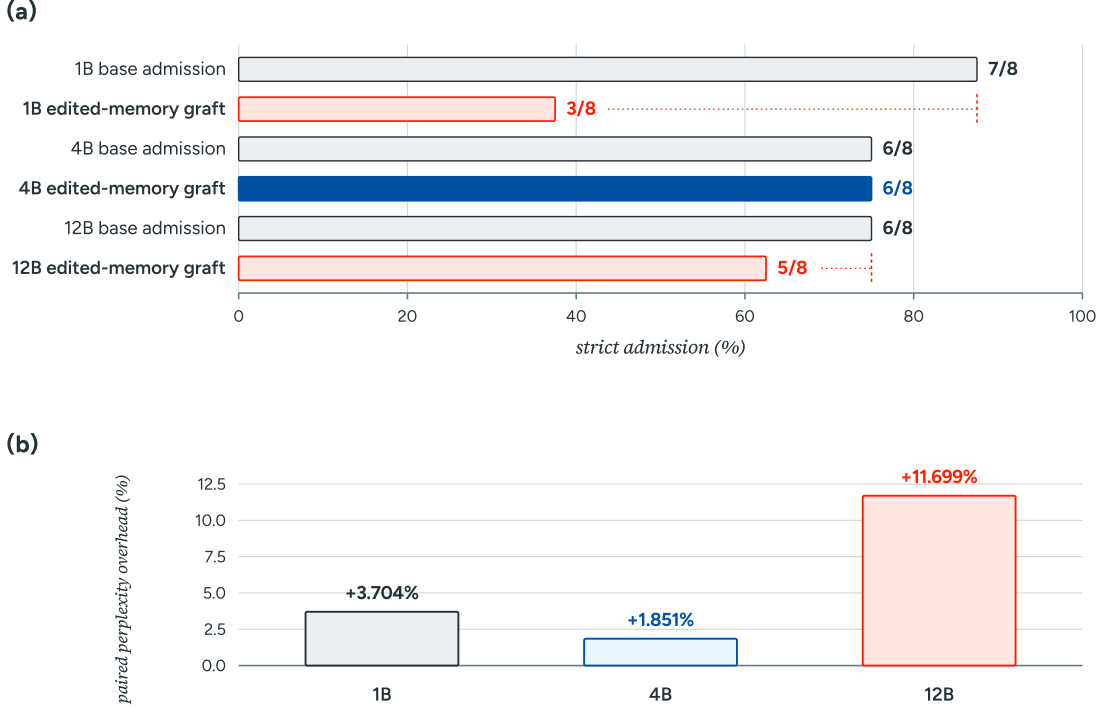


Figure 4: Admission and quality cost of the frozen configuration at three scales. (a) Strict admission, where 4B is the only tested scale that preserves the base model’s eligibility. (b) Paired perplexity overhead, where 12B has the largest measured quality cost. A dashed marker indicates that the graft no longer matches its own base admission. The three checkpoints are sensitivity outcomes and were not selected by any model-selection procedure.

We checked both agreement with the refit and completion of the decrement at the two certified scales. At 1B, the median and maximum KL against the refit on retained keys are $1.86 \times 10^{-13}/1.25 \times 10^{-10}$; at 4B they are $2.36 \times 10^{-12}/6.48 \times 10^{-11}$. No probe exceeds 10^{-6} . The reverse decrement failed its checks in 396/640 1B and 2,020/3,040 4B attempted affected solves. Refitting completed each of those updates, so every deletion still reached its reference. These denominators include affected solves only; solve points before the deleted position do not need an update. Scoring the already-constructed exact, refit, and decay states takes 561.5 s for 32 probes at 1B and 2,354.6 s at 4B; these are the costs of scoring the audit and do not measure deletion latency.

B.2 Behavioral attack details

At $k = 1$, exact-phras leakage is 0.28% edited and 0.42% never-stored. At $k = 200$, each condition leaks on 3/18 field queries (16.7%, the cohort rate), with paired record-bootstrap difference 0.0 [0.0, 0.0]. Present and prompt-only are 18/18 (100%), and decay is 5/18 (27.8%). To check reproducibility, we repeated the sampling arm with the same seeds, decoding settings, and batch size. It reproduced every $k = 200$ endpoint exactly (18/18, 18/18, 3/18, 3/18, and 5/18 for present, prompt-only, edited, never stored, and decay) and the $k = 1$ rates to within 0.02 percentage points, while individual sampled completions differed. Thus the per-query endpoints reproduced on this hardware even though the per-sample counts were not bit-exact. Elicitation, related-data updates, and LiRA are described in the released attack aggregate.

For LiRA AUC intervals, we resampled records with replacement and took percentile endpoints over 10^4 draws: six-record cohort 0.534 [0.513, 0.601], broad cohort 0.517 [0.516, 0.539]; AUCs computed

separately for each record span 0.47–0.76 and 0.53–0.73. The broad LiRA policy includes 16/512 edited test deletions routed to raw repack. Replacing those scores with arbitrary values moves the AUC anywhere in $[0.486, 0.549]$; we report that range as a sensitivity check, distinct from a confidence interval. The frozen interval code sorts the 10,000 pooled AUCs and selects ordered entries 251 and 9,750 (one-based), yielding the asymmetric broad interval $[0.5157394409179688, 0.5387916564941406]$. It also reports the upper central AUC computed separately for each record for even sample sizes. Table 7 keeps these conventions and rounds only for display.

C LongMemEval protocols and complete counts

Section 9 reports the findings of the two LongMemEval studies. This appendix gives their protocols and the complete counts behind each reported rate, beginning with the selected examples and then describing the two cohorts.

C.1 Selected decoded examples and their limits

The mortgage, screwdriver, and tennis examples are post hoc illustrations from the completed 96-history study, not additional experiments. Each target has three nested suffix variants assembled from the pinned public LongMemEval oracle, with the same target and retained control. The protocol deletes the selected latest user–assistant round, preserves earlier evidence literally, and leaves later retained turns unchanged. A fresh raw-history reference re-encodes those retained inputs; it does not regenerate them. For each probe, the model receives a registered user turn containing `Current Date:`, the official question, and “Reply with only the remembered value, without explanation.” A model turn then begins. Repeated deterministic decodes are reproducibility checks, not independent samples. Solver checks target the fixed- C problem on the retained keys; no separate refit response was decoded.

Mortgage update. The official target question is quoted in Section 9.1, with query date December 9, 2023. The owned November 30 round occupies token positions $[876, 1233)$ in each assembled history. The earlier August 11 round mentions pre-approval for \$350,000; the removed round mentions \$400,000. The retained question, dated May 27, 2023, is “I wanted to follow up on our previous conversation about YouTube videos for workplace posture. Can you remind me of the Mayo Clinic video you recommended?” The saved benchmark gold contains the video title “How to Sit Properly at a Desk to Avoid Back Pain” and URL <https://www.youtube.com/watch?v=UfOvNlX9Hh0>.

Table 10: All three mortgage variants. Entries are saved generated answers to the target query; the video URL is present under all four conditions in every variant. Variant 0’s edited answer also includes the video title.

variant	tokens	after edit span	present	prompt	local edit	rebuild
0	2,596	1,363	\$400,000	\$400,000	\$350,000	\$350,000
1	2,764	1,531	\$400,000	\$400,000	\$350,000	\$350,000
2	2,941	1,708	\$400,000	\$400,000	\$350,000	\$350,000

The local policy used refit fallback in 480/560, 508/600, and 534/640 affected head/solve-point solves, respectively. The high use of refitting leaves the speed advantage of a local edit unestablished. The retained URL matches the broader endpoint in all variants, although the registered whole-answer substring endpoint does not match the complete gold sentence. Aggregate retained matching is reported in Section 9.1.

A local edit and rebuild can answer differently. For the screwdriver target, the earlier exchange says the small screwdriver was misplaced. The latest user says “I actually have a spare screwdriver”; the

assistant responds “Spare screwdriver: Ah, excellent!” [excerpts]. The deleted user–assistant round occupies [1189, 1594]. The official query is “Do I have a spare screwdriver for opening up my laptop?” Table 11 preserves the suffix-dependent outcome. All conditions return “38” to an unrelated numerical control whose saved gold is “38 subjects”; neither existing retained matcher marks those answers correct. The reported labels follow those registered matchers.

Table 11: All three screwdriver variants. Editing and rebuilding disagree in variant 0; the other two rebuilt answers also contain the target “Yes”.

variant	tokens	present	prompt	local edit	rebuild
0	3,160	Yes	No	Yes	No
1	3,290	Spare screwdriver	Yes	Yes	Yes
2	3,038	Yes	No.	Yes	Yes

The local policy used refit fallback in 538/600, 564/640, and 510/560 affected solves. These examples show a difference in decoded behavior; they do not identify which surviving row or later turn caused it.

A harder question can expose an endpoint limitation. The tennis target asks both previous and current frequency. Its full gold answer says previously every week on Sunday and currently every other week on Sunday. Across histories of 2,339, 2,356, and 2,674 tokens, present and prompt-only conditions generate only “Every other week”, whereas editing and rebuilding generate “Weekly”. The retained answer is *MusicTheory.net* throughout. Neither partial target answer matches the full registered gold, so matcher-negative output alone does not establish successful forgetting. The frozen counts retain the registered matcher labels for these incomplete answers.

The source question IDs are 852ce960 (mortgage), 42ec0761 (screwdriver), and f685340e (tennis). The source-free cohort manifest fixes all three history IDs and token spans per target. The response audit connects the saved outputs to those histories, and a local review manifest records their artifact paths and hashes.

C.2 Bicycle and Speyer case: teacher-forced scores

The bicycle example in the introduction comes from the 16-history pilot. Its 3,103-token assembled history contains an earlier three-bike discussion; the deleted round reports the new hybrid bike, updates the count to four, and includes the assistant’s echo. The retained exchange supplies the Speyer phone number. We use teacher-forced scoring to see how deletion changes the probabilities of the saved answers: the model receives each saved answer token as it proceeds, and we record its probability. For “How many bikes do I currently own?”, the one-token gold “4” moves from rank 2 to rank 28 after editing. The 22-token phone answer keeps rank 1 at its first token; its mean log probability changes from -2.31560×10^{-5} to -4.64238×10^{-5} nats per answer token. Because the previous gold tokens are supplied during scoring, the phone-number result does not tell us whether the model would generate the complete number on its own. The maximum local output KL is 4.32788×10^{-17} nats across two probes, in the direction from the edited policy to the fixed-*C* refit. The bicycle probe’s distinct KL from the raw rebuild to the edited policy is 0.00176439 nats over the first-token vocabulary. Of 560 affected solves, 522 used exact-refit fallback and 38 completed incrementally. The record ID ends in c8276e265e3db489c090cdcc.

C.3 Teacher-forced pilot on sixteen histories

We used the pilot to examine how the probabilities of two saved answers change when one exchange is removed and the other retained. LongMemEval evaluates memory over timestamped, multi-session chat histories [19]. We adapted it to Gemma-3-4B-IT, a different checkpoint from the pretrained ones in

the utility study; this adapted test does not produce a leaderboard score. The owned unit includes every complete user–assistant round that carries the answer in the latest target session, and a disjoint record supplies the retained probe. We froze the ordered 16-history cohort with its questions, answers, controls, and probes before any model output was seen.

A history needs enough distance from the deleted exchange to test memory outside the local window, as well as enough surviving rows to satisfy the fixed- C feasibility requirement. A deterministic pass over the frozen source inventory appended complete rounds where needed to meet these requirements. It changed no record, question, answer, control, or probe. All 16 histories remain in the denominator: 10 recall the target, all 16 expose the retained control, and 10 satisfy both gates and form the subset that Table 12 reports.

Every history required exact refit at one or more affected solves, so every deletion in the pilot was classified as a refit (522 of 560 solves in the case of Appendix C.2). None required a raw repack. These counts identify the executed route; they do not measure its speed. We compare the edited policy with an independent fixed- C refit on 32 registered probes over all 16 histories, with mean 9.59×10^{-18} and maximum 1.75×10^{-16} nats against a declared tolerance of 10^{-6} . Section 9 reports the separate comparison with the never-stored state.

The LongMemEval revision, the Gemma tokenizer and model revision, the ordered record bindings, the geometry policy, the admission report, the method lock, and the final report are pinned by hash in the release, and the released report stores neither conversation text nor full-vocabulary vectors. Table 12 lists every condition of the pilot against its two comparators: the behavioral distance runs from the never-stored state to each condition, and the certificate runs from the edited policy to the fixed- C refit.

Table 12: Teacher-forced pilot conditions on the 10-history jointly admitted subset. Suppression is the decrease in the target answer’s mean log probability relative to the record-present run, in nats. Target and retained KL are full-vocabulary first-token KL from the never-stored state to the condition. The three constructions of the edited state agree with one another, which is the certificate (tolerance 10^{-6}), and their agreement is a separate quantity from their distance to the never-stored state.

condition	target suppression	target KL from never-stored	retained KL from never-stored
present control	0.00	9.03	3.37×10^{-5}
never-stored state (rebuild)	6.57	0.00	0.00
<i>three constructions of the same edited state; their agreement is the certificate</i>			
edited policy (fixed- C decrement/refit)	8.33	2.15	1.90×10^{-6}
fixed- C refit on retained keys	8.33	2.15	1.90×10^{-6}
FP32 masked-refit proxy	8.34	2.15	1.89×10^{-6}
coefficient decay ($\gamma = 0.01$)	6.30	0.30	7.23×10^{-5}
target-free prompt suppression	0.53	6.03	4.98×10^{-5}

C.4 Decoded follow-up: cohort and decoding

The follow-up tests what the model actually generates on $K = 32$ target clusters and $n = 96$ histories, separate from the pilot above (Appendix Table 9). We froze the 32 clusters first and packed three histories per cluster independently. Every history completed under four conditions: record present, local deletion, rebuilding from the retained history, and a prompt-only instruction to forget. This produced 384 decoded outputs. We kept histories even when the model failed to recall the record, so every rate below uses all 96 histories. Decoding each probe greedily twice produced identical text and token identifiers, checking deterministic reproducibility rather than variation across sampled answers.

C.5 Screening for disclosure: a matcher, a judge, and two reviewers

We screened the 384 decoded outputs in three stages. First, a deterministic matcher compared each output with the source answer under five checks applied together: a casefolded substring test, a test

after punctuation and articles are stripped, a numeric and unit test, a URL test, and a named-entity test. An output that passed any of them counts as a disclosure. There were 131 of these, and we did not review them further. Second, a language model judge (GPT-5.6) read every one of the 253 outputs the matcher had cleared, seeing the question, the reference answer, and the candidate response but neither the condition nor the matcher’s verdict, and returned one of three labels: disclosure, no disclosure, or ambiguous. Every request returned a valid response. Because the provider serves the judge under a name that can change between calls, the local audit retains the requested and returned model names along with the frozen prompt, response schema, and every request and response.

Third, two reviewers read a sample of the outputs, blinded to condition. The sample held the 19 outputs the judge had flagged and 19 outputs it had cleared, the latter chosen by hash so that the choice could not depend on their content (38 outputs in all). The first reviewer labeled every output in the sample, and the second read the flagged outputs the first had not labeled as disclosures (3 distinct texts) and agreed with the first on each. The 38 outputs contain only 21 distinct question, answer, and response triples, because the same response text recurs across conditions. We count outputs below, but the duplicate texts are not independent judgments.

Of the 19 flagged outputs, the reviewers labeled 15 as disclosures, 3 as no disclosure, and 1 as ambiguous, and they labeled all 19 controls as no disclosure. Every confirmed disclosure was a form of the answer that the matcher had failed to recognize: 7 were normalization failures and 8 were an alias or an inflected form of the answer, and none was a paraphrase. Most reviewed outputs were selected because the judge had flagged them. The review therefore characterizes those flags and a control sample, rather than estimating how often each condition discloses. The 2 outputs the judge itself marked ambiguous were not reviewed.

C.6 Disclosure rates and the paired comparison

For the endpoint an output counts as a disclosure when the matcher flagged it or the reviewers labeled it one; for flagged outputs the reviewers did not read, the judge’s label stands. Ambiguous labels give a lower and an upper count. The reviewed labels changed the aggregate counts as follows. After review, the 253 outputs the matcher cleared hold 15 disclosures, 235 non-disclosures, and 3 ambiguous outputs, against the judge’s 19, 232, and 2 before review. Per history, the edited policy disclosed the deleted answer in 15/96, the never-stored rebuild in 13/96, the record-present run in 64–66 of 96, and the prompt-only instruction in 54–55. The paired edited-minus-rebuild difference is $+2/96 = +2.1$ percentage points, with a 95% interval of $[0.0, 5.2]$ (a bootstrap over the 32 clusters), and the prompt-minus-edited difference is at least 40.6 points (interval $[17.7, 62.5]$). These intervals hold the labels fixed and do not include uncertainty from selecting the control outputs or from the reviewers’ judgments. The $+2/96$ difference does not resolve equivalence, because we set no equivalence margin in advance. Against margins chosen afterwards, the upper end of the interval, 5.2 points, misses a 5-point margin and meets a 10-point one. The counts put the edit well below the prompt instruction and within a few points of rebuilding on this probe, with disclosures remaining under both procedures.

C.7 Matching responses and correct responses

The edited policy returned exactly the same target response as the never-stored rebuild in 77 of 96 histories, and the same response under at least one of the matcher’s normalizations in 87. To assess retained memory, we also compared responses with their saved answers. On the retained probe, the answer was correct under the matcher in 46/96 histories with the record present, 46/96 after the rebuild, and 44/96 under the edited policy, and the edited policy’s retained response matched the record-present response under some normalization in 82/96. This comparison matters because the edit and rebuild can agree on an incorrect answer.

Two histories illustrate why response agreement and correctness need separate measurements. We chose one from each suffix stratum by SHA-256 rank before reading any response and kept every condition for each. In the history whose suffix had no lexical contact with the deleted answer, the gold

target was “38,” but every condition answered “37,” including the record-present control. The retained answer, “Three,” was stable throughout. Deletion cannot uniquely explain a target error that was already present in the control. In the history with lexical contact, the record-present and prompt-only conditions answered “5,” while the rebuild and edit answered “Four.” Here the edit tracked the rebuild, but every condition missed the paired retained restaurant. Both histories remain in the original cohort and in the rates reported above.

C.8 The frozen suffix

We examined the surviving suffix because later text can carry information from a deleted exchange back into a rebuild. Copies can appear as an assistant’s echo, a derived value, or a summary; the delta-attention study gives a worked example [13]. We scanned the turns after each deleted record for lexical contact with the target answer and found it in 24 of 96 histories, spanning 10 of 32 clusters. The suffix was assembled from independent support sources and was never generated downstream of the target, so contact marks possible contamination and does not establish causal dependence. The scanner catches explicit restatements and some normalized forms, generally misses derived traces and summaries, and can match short or numeric strings. The detected count is neither a lower nor an upper bound on contamination.

We assessed whether lexical contact was associated with the edited policy’s residual disclosures (the 15 of 96 in Section 9). Among the 24 histories with contact, 1 disclosed and 23 did not; among the 72 without contact, 14 disclosed and 58 did not, for disclosure rates of 4.2% and 19.4% (Fisher’s exact test, two-sided, $p = 0.105$). Disclosure was less frequent in the lexical-contact stratum. The matcher missed 1 edited-policy output that the judge then flagged, and the reviewers confirmed it, so the cross-tabulation is unchanged from the judge’s labels. Lexical contact therefore does not explain the residual disclosure in this study. A broader deletion request could also ask for later turns to be regenerated if they were caused by the deleted record. Our frozen-history comparison does not test that operation; it is discussed in the delta-attention study [13].